Xiaoling Hu

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Current Position

- **Harvard Medical School, Harvard University, USA** Aug. 2023 - Present
  Postdoctoral Research Fellow
  - Hosted by Prof. Juan Eugenio Iglesias and Prof. Bruce Fischl

Research Interests

My research interests lie in machine learning, computer vision, with applications in healthcare, multimodal AI, generative AI, VLM/LLMs, and trustworthy learning systems. In particular, I am interested in:

- **Topological Deep Learning**
- **Trustworthy Machine Learning**
- **Multimodal AI and Generative AI (GenAI)**
- **VLM/LLM Reasoning, Efficiency and Confidence**

Education

- **Department of Computer Science, Stony Brook University, USA**
  Jan. 2018 - Jun. 2023
  Ph.D. in Computer Science
  - Advisor: Chao Chen
  – Thesis: Learning Topological Representations for Deep Image Understanding

- **Department of Electronic Engineering, Tsinghua University, China**
  Sep. 2014 - Jun. 2017
  Master of Engineering

- **Department of Electronic Engineering, Huazhong University of Science and Technology, China** Sep. 2010 - Jun. 2014
  Bachelor of Science

Selected Honors and Awards

- Top Area Chair, NeurIPS 2026
- Oral, CVPR (Top 1%) 2025
- Catacosinos Fellowship, Stony Brook University 2023
- Spotlight, ICLR (Top 6%) 2023
- Oral, ECCV (Top 3%) 2022
- Spotlight, ICLR (Top 6%) 2021

Professional Service

Area Chair

- Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
- International Conference on Learning Representations (ICLR) 2027
- International Conference on Machine Learning (ICML) 2027
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, 2027
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2026, 2027

- AAAI Conference on Artificial Intelligence (AAAI) 2027
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026

Workshop and Tutorial Organizer

- 8th Workshop on *GRaphs in biomedicAl Image anaLysis (GRAIL)*, MICCAI 2026
- Workshop on *Topology- and Graph-Informed Imaging Informatics (TGI3)*, MICCAI 2025
- Workshop on *The First Workshop on Topology- and Graph-Informed Imaging Informatics (TGI3)*, MICCAI 2024
- Tutorial on *Topology-Driven Image Analysis*, MICCAI 2023

Seminar Organizer

- Co-organizer, CML Seminar Series, Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School

Reviewer

- *Conferences*: NeurIPS, ICLR, ICML, CVPR, ICCV, ECCV, AAAI, AISTATS, WACV, LoG, MIDL, MICCAI
- *Journals*: TPAMI, TMLR, IEEE TMI, Medical Image Analysis (MedIA)

Selected Publications

(* indicates equal contribution, __ denotes students I (co-)mentored on that paper, ‡ denotes (co)-senior supervision)

[1] **Topology-Aware Uncertainty for Efficient Vessel Annotation**
Saumya Gupta, Sumedh Ghavat, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
*IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), under major revision*

[2] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**
Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta, Jiawei Zhou, Chao Chen
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)*, 2027

[3] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**
Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias
*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2026

[4] ***RankByGene*: Gene-Guided Histopathology Representation Learning Through Cross-Modal Ranking Consistency**
Wentao Huang, Meilong Xu, **Xiaoling Hu**, Shahira Abousamra, Aniruddha Ganguly, Saarthak Kapse, Alisa Yurovsky, Prateek Prasanna, Tahsin Kurc, Joel Saltz, Michael L. Miller, Chao Chen
*IEEE Transactions on Medical Imaging (**TMI**)*, 2026

[5] **Multi-Channel Uncertainty-Weighted Score Matching for Conditional Diffusion in Medical UDA**

Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen

*European Conference on Computer Vision (**ECCV**)*, 2026

[6] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**

Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026

[7] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**

Meilong Xu*, **Xiaoling Hu***, Shahira Abousamra, Chen Li, Chao Chen

*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2025

*Short version selected as an **Oral Presentation** at the Imageomics Workshop at NeurIPS 2025*

[8] ***Learn2Synth*: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**

Xiaoling Hu, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl‡, Yaël Balbastre‡

*IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025

[9] ***TopoCellGen*: Generating Histopathology Cell Topology with a Diffusion Model**

Meilong Xu, Saumya Gupta, **Xiaoling Hu**, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2025 (**Oral Presentation, top 1% of submissions**)

Short version also presented at CVPR 2025 Workshop GMCV

[10] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**

Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti‡, Juan Eugenio Iglesias‡

*International Conference on Learning Representations (**ICLR**)*, 2025

[11] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**

Meilong Xu, **Xiaoling Hu**, Saumya Gupta, Shahira Abousamra, Chao Chen

*European Conference on Computer Vision (**ECCV**)*, 2024

[12] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**

Peirong Liu, Oula Puonti, **Xiaoling Hu**, Daniel C. Alexander, Juan Eugenio Iglesias

*European Conference on Computer Vision (**ECCV**)*, 2024

[13] **Semi-Supervised Contrastive VAE for Disentanglement of Digital Pathology Images**

Mahmudul Hasan, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Joel Saltz, Chao Chen

*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[14] **Hard Negative Sample Mining for Whole Slide Image Classification**
Wentao Huang, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Chao Chen
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[15] **Spatial Diffusion for Cell Layout Generation**
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Meilong Xu, Chao Chen
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[16] **Anomaly-Guided Weakly Supervised Lesion Segmentation on Retinal OCT Images**
Jiaqi Yang, Nitish Mehta, Gozde Merve Demirci, **Xiaoling Hu**, Meera Ramakrishnan, Mina Naguib, Chao Chen, Chialing Tsai
*Medical Image Analysis (**MedIA**)*, 2024

[17] **Topology-Aware Uncertainty for Image Segmentation**
Saumya Gupta, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2023

[18] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Chao Chen
*IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2023

[19] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**
Aishik Konwer, **Xiaoling Hu**, Xuan Xu, Joseph Bae, Chao Chen, Prateek Prasanna
*IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2023

[20] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**
Xiaoling Hu, Dimitris Samaras, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2023 (**Spotlight Presentation, top 6% of submissions**)
*Short version selected as an **Oral Presentation** at Medical Imaging meets NeurIPS Workshop, 2022*

[21] **Confidence Estimation Using Unlabeled Data**
Chen Li, **Xiaoling Hu**, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2023

[22] **Structure-Aware Image Segmentation with Homotopy Warping**
Xiaoling Hu
*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2022

[23] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**
Saumya Gupta*, **Xiaoling Hu***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tassiopoulos, Prateek Prasanna, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2022 (**Oral Presentation, top 3% of submissions**)

[24] **Trigger Hunting with a Topological Prior for Trojan Detection**
Xiaoling Hu, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2022

[25] **A Manifold View of Adversarial Risk**
Wenjia Zhang, Yikai Zhang, **Xiaoling Hu**, Mayank Goswami, Chao Chen, Dimitris Metaxas
*International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, 2022

[26] **Topology-Attention ConvLSTM Network for 3D Image Segmentation**
Jiaqi Yang*, **Xiaoling Hu***, Chao Chen, Chialing Tsai
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2021

[27] **Topology-Aware Segmentation Using Discrete Morse Theory**
Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2021 (**Spotlight Presentation, top 6% of submissions**)

[28] **Topology-Preserving Deep Image Segmentation**
Xiaoling Hu, Li Fuxin, Dimitris Samaras, Chao Chen
*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2019

Preprints

[1] **TriSegAgent: 3D Medical Annotation with VLM Agents via Tri-View Context Compression**
Qingqiao Hu*, Jialin Li*, Meilong Xu, Kehan Qi, Yipeng Zhang, Bran Hongwei Li, Klaus Mueller‡, **Xiaoling Hu**‡
Tech Report, 2026

[2] **ORCE: Order-Aware Alignment of Verbalized Confidence in Large Language Models**
Chen Li, **Xiaoling Hu**, Songzhu Zheng, Jiawei Zhou, Chao Chen
Tech Report, 2026

[3] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**
Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu, Kehan Qi, Dimitris Samaras, Chao Chen
Tech Report, 2026

[4] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**
Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen
Tech Report, 2025

[5] **RB-FT: Rationale-Bootstrapped Fine-Tuning for Video Classification**
Meilong Xu, Di Fu, Jiaxing Zhang, Gong Yu, Jiayu Zheng, **Xiaoling Hu**, Dongdi Zhao, Feiyang Li, Chao Chen, Yong Cao
Tech Report, 2025

[6] **Learning to Upscale 3D Segmentations in Neuroimaging**
Xiaoling Hu, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias
Tech Report, 2025

Mentoring

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24, ECCV'26**), Ph.D. student at Department of BMI, Stony Brook University → Incoming Postdoc at Stanford University
  Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Wentao Huang (**MICCAI'24, CVPR'26, TMI'26**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Qingqiao Hu (**WACV'27**), Ph.D. student at Department of CS, Stony Brook University
  Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University
  Since Fall 2025
- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University
  Summer 2023 - Summer 2024
- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University
  Fall 2021 - Summer 2023
- Jiaqi Yang (**ISBI'21, MICCAI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY
  Spring 2020 - Summer 2023
- Luca Drole, Master's student in BME, ETH Zurich
  Spring 2025 - Fall 2025
- John Xie, high school student → University of Michigan
  Summer 2021

Invited Talks

Principled Learning for Medical AI: Structure, Reliability, and Interpretability

- Kahlert School of Computing & SCI Institute, University of Utah — Mar. 2026
- School of Computing, University of Georgia — Mar. 2026
- Department of EECS, University of Tennessee, Knoxville — Feb. 2026
- Department of CS, Missouri S&T — Jan. 2026
- Department of CSE, University of North Texas — Nov. 2025
- HIT Webinar - *Healthcare, Intelligence, Technology* — Aug. 2025

***Learn2Synth*: A Learnable Data Synthesis Strategy for Image Segmentation**

- Nobrainer Seminar, Massachusetts Institute of Technology — Jun. 2024

Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University — Feb. 2024

Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver — Oct. 2023

- Geometry and Topology meet Data Analysis and Machine Learning Aug. 2021

Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University Apr. 2023
- Department of BMI, Ohio State University Mar. 2023
- Department of CS, Rochester Institute of Technology Feb. 2023
- Department of ECE, University of California, Riverside Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School Nov. 2022

Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans Dec. 2022

Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute Oct. 2022

Topology-Aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) May 2021

Industry Experience

- **Allen Institute, USA** May 2022 - Aug. 2022
  Research Intern
  Mentor: *Dr.* Matheus Viana
  Topic: Topology-Aware Image Segmentation
- **United Imaging Intelligence (UII), USA** May 2021 - Aug. 2021
  Research Intern
  Mentor: *Dr.* Shanhui Sun
  Topic: Deep Shape Model Based Network
- **Tencent Youtu Lab, China** Jun. 2017 - Jan. 2018
  Research Intern
  Mentor: *Dr.* Yuwing Tai
  Topic: Clothes Detection, Attribute Prediction